

1. Agar $n^3 = 48 \cdot m$ va $m, n \in N$ bo'lsa, $m + n$ yig'indining eng kichik qiymatini toping.

- A) 24
- B) 36
- C) 48
- D) 54

2. Hisoblang:

$$\frac{12\frac{4}{5} \cdot 3\frac{3}{4} - 4\frac{4}{11} \cdot 4,125}{2\frac{4}{7} : \frac{3}{35}}$$

- A) 1
- B) 2
- C) 3
- D) 4

3. Ikki velosipedchi oralaridagi masofa 28 km bo'lgan A va B punktlardan qarama-qarshi yo'lga chiqdi. 1 soatdan so'ng uchrashib, shu tezlik bilan yo'lda davom etishdi.

1-velosipedchi B punktga 35 minut oldin keldi. Velosipedchilar tezliklarini toping.

- A) 16; 12
- B) 14; 18
- C) 12; 20
- D) 12; 18

4. Ikkita usta detallarni yasashi kerak edi. Uch soat birga ishlagandan so'ng, ikkinchi usta yana 4 soat yolg'iz ishladi. Shundan so'ng, vazifa 12,5% ko'proq bajarilganligi ma'lum bo'ldi. Agar vazifani alohida-alohida bajarganda, ikkinchi usta birinchisiga qaraganda 4 soat kamroq vaqt sarflaydi. Ikkinchi usta alohida ishlasa qancha vaqt sarflaydi?

- A) 12
- B) 10
- C) 8
- D) 15

5. Ifodani soddalashtiring:

$$\frac{x^2 - y^2}{x + y} - \frac{x^3 - y^3}{x^2 + xy + y^2}$$

- A) $\frac{x+y}{x-y}$
- B) $\frac{x-y}{x+y}$
- C) $\frac{2xy}{x+y}$
- D) 0

6. Kasrni qisqartiring:

$$\frac{(\sqrt[3]{9} + \sqrt{3})^2}{\sqrt[3]{3} + 2 \cdot \sqrt[6]{3} + 1}$$

- A) 3
- B) 2
- C) $\frac{1}{3}$
- D) $\frac{1}{2}$

7. Soddalashtiring:

$$\left(\frac{\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}} - (\sqrt{x} + \sqrt{y}) \cdot (\sqrt{x} - \sqrt{y})^{-1} \right) \cdot \frac{\sqrt{x} - \sqrt{y}}{3(\sqrt{y})^3}$$

- A) $\frac{1}{\sqrt{y}}$
- B) $-\frac{1}{2\sqrt{y}}$
- C) $\frac{1}{y}$
- D) $-\frac{1}{y}$

8. a_1 ; a_2 va a_3 arifmetik progressiyaning ketma-ket hadlari, a_1^2 ; a_2^2 va a_3^2 lar esa berilgan tartibda geometrik progressiyaning ketma ket hadlari bo'lsa, geometrik progressiya maxrajini toping.

- A) $\frac{1}{2}$
- B) 1
- C) $\frac{1}{3}$
- D) aniqlab bo'lmaydi

9. 6 ta haddan iborat bo'lgan geometrik progressiyada dastlabki uchta hadi yig'indisi 14 ga, oxirgi uchta hadi yig'indisi 112 ga teng bo'lsa, maxrajini toping.

- A) $\frac{1}{2}$
- B) 2
- C) $\frac{1}{3}$
- D) 3

10. $a = 2024$, $b = -2023$ bo'lsa, $a^3 + b^3 + a^2 + b^2 + 1 + 5ab$ ning qiymatini toping.

- A) 1
- B) -1
- C) 2
- D) 3

11. Ifodani soddalashtiring:

$$\left(\frac{3}{3-a} - \frac{a}{3+a}\right) : \left(\frac{3-a}{3} - \frac{3+a}{a}\right) : \frac{a}{9-a^2}$$

- A) 3
- B) -3
- C) 9
- D) -9

12. $2024^\circ - 2023' - 2022''$ ni gradus, minut sekunda ifodalang.

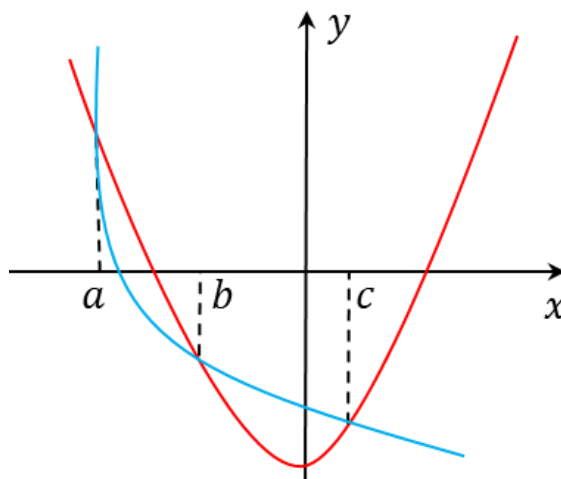
- A) $1990^\circ 42' 18''$
- B) $1989^\circ 42' 18''$
- C) $1989^\circ 43' 18''$
- D) $1990^\circ 43' 18''$

13. Ifodani soddalashtiring;

$$\frac{\operatorname{tg} \frac{\alpha}{2} - \operatorname{ctg} \frac{\alpha}{2}}{\operatorname{tg} \frac{\alpha}{2} + \operatorname{ctg} \frac{\alpha}{2}}$$

- A) $\sin \alpha$
- B) $\cos \alpha$
- C) $-\sin \alpha$
- D) $-\cos \alpha$

14. Rasmda $y = f(x)$ parabola va $y = g(x)$ ko'rsatkichli funksiya grafiklari tasvirlangan. $f(x) > g(x)$ tengsizlikning yechimlar to'plamini toping.



- A) $(-\infty; a) \cup (c; \infty)$
- B) $(b; c)$
- C) $(a; c)$
- D) $(a; b) \cup (c; \infty)$

15. $\sqrt{\log_3 x^9} - 4 \log_9 \sqrt{3x} = 1$ tenglamaning haqiqiy ildizlari ko'paytmasini toping. (Agar u bitta bo'lsa, shu ildizini toping)

- A) 3
- B) 27
- C) 81
- D) 243

16. Tenglamaning butun ildizlari yig'indisini hisoblang.

$$|x - 2| + |x + 3| = 5$$

- A) -3
- B) -1
- C) -6
- D) 6

17. Agar x_1 va x_2 sonlar $2024x^2 - 2025x + 1 = 0$ tenglamaning ildizlari bo'lsa,

$\left(x_1 + \frac{1}{x_1 x_2}\right) \left(x_2 + \frac{1}{x_1 x_2}\right) - \frac{1}{x_1^2 x_2^2}$ ni hisoblang.

- A) $1012 \frac{1}{2024}$
- B) $2024 \frac{1}{2024}$
- C) $2025 \frac{1}{2024}$
- D) $1013 \frac{1}{2025}$

18. Tengsizlikni yeching:

$$(8x + 3)\sqrt{x^2 - 6x + 8} \geq 0$$

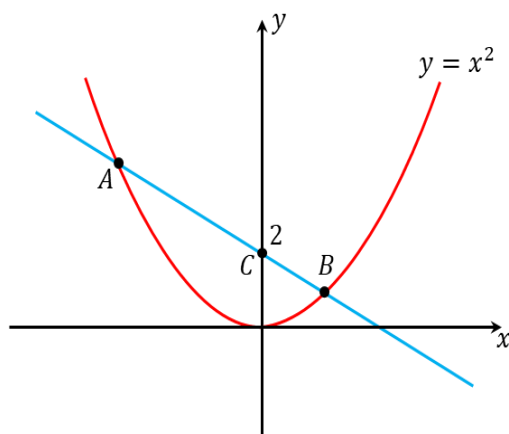
- A) $\left[-\frac{3}{8}; \infty\right)$
- B) $(-\infty; 2] \cup [4; \infty)$
- C) $\left[-\frac{3}{8}; 2\right] \cup [4; \infty)$
- D) $\left[-\frac{3}{8}; 2\right]$

19. Tengsizlikni yeching;

$$\frac{x-2}{x} - \frac{x+2}{x-2} < 2$$

- A) $(2; \infty)$
- B) $(-2; 0) \cup (0; 2)$
- C) $(-\infty; -2) \cup (0; 1) \cup (2; \infty)$
- D) $(-\infty; 0) \cup (2; \infty)$

20. Rasmda parabola va to'g'ri chiziq grafiklari tasvirlangan. $A(n; n^2)$ va $|AC| = 3|BC|$ tenglik o'rinli bo'lsa, $n^2 = ?$



- A) 6
- B) 7
- C) 8
- D) 9

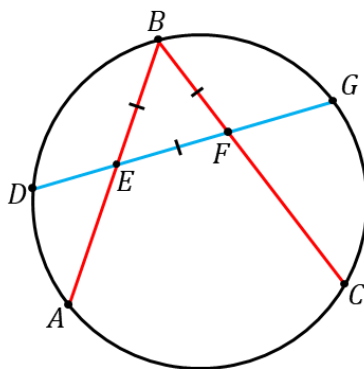
21. Aniqmas integralni hisoblang: $\int \frac{\sqrt{1+\ln x}}{x} dx$

- A) $\sqrt{(1 + \ln x)^3} + C$
- B) $\frac{2}{3}\sqrt{(1 + \ln x)^3} + C$
- C) $\frac{1}{3}\sqrt{(1 + \ln x)^3} + C$
- D) $2\sqrt{(1 + \ln x)^3} + C$

22. $f(x) = 2024^x \ln 2024x$ bo'lsa, $f'(1)$ ni toping.

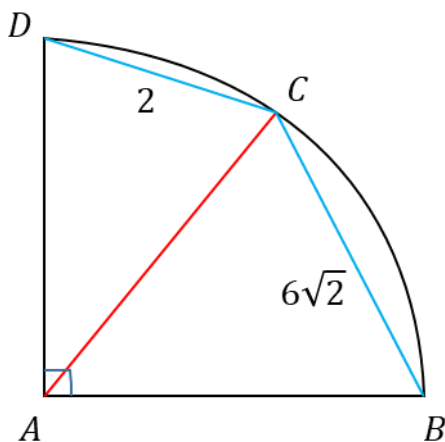
- A) $2 \ln 2024 + 2024$
- B) $4048 \ln 2024 + 1$
- C) $2024(\ln^2 2024 + 1)$
- D) $2024(2 \ln 2024 + 1)$

23. Aylananing B nuqtasidan ikkita BA va BC vatar o'tkazildi. BA va BC vatarni mos ravishda E va F nuqtda kesishuvchi DG vatar o'tkazildi(rasm). Bunda $BE = EF = BF$, $DE = 2$, $AE = 3$, $FG = 4$ bo'lsa, FC kesma uzunligini toping.



- A) 4
- B) 5
- C) 6
- D) 7

24. Rasmdagi chizmada markazi A nuqtada bo'lgan aylananing DC vatari 2 ga va CB vatari $6\sqrt{2}$ ga teng bo'lsa, ABC uchburchakning yuzini toping.

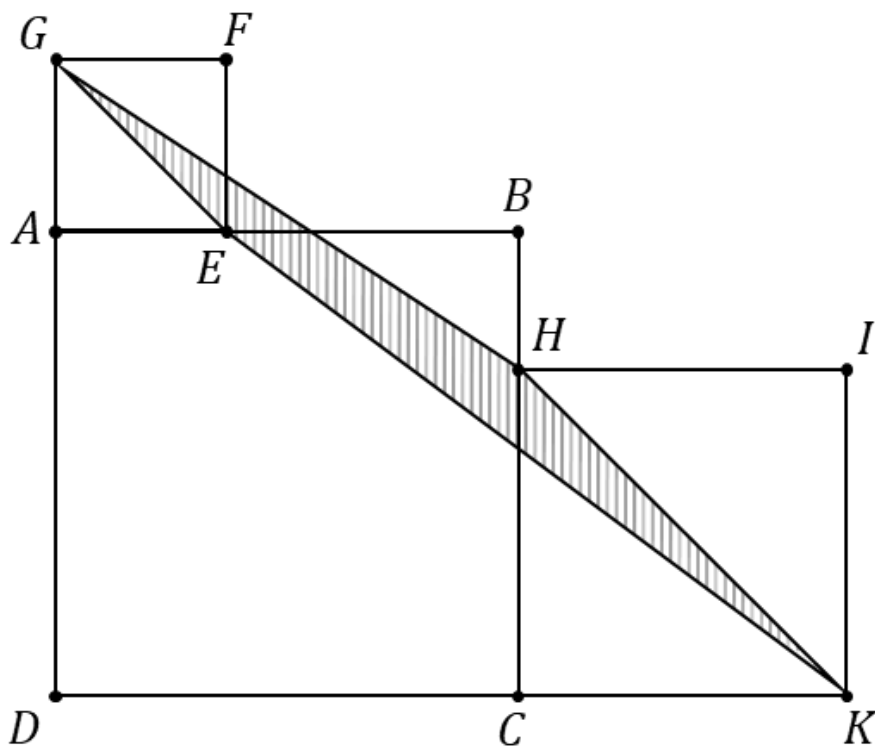


- A) 24
- B) 6
- C) 7
- D) 31

25. To'g'ri burchakli trapetsiya asoslari 3 va 4 ga teng bo'lsa, unga ichki chizilgan aylana radiusini toping.

- A) $\frac{7}{12}$
- B) $\frac{13}{7}$
- C) $\frac{12}{7}$
- D) $\frac{7}{13}$

26. Rasmda $ABCD$, $CHIK$ va $AGFE$ kvadratlar tasvirlangan. Bu kvadratlar yuzalari mos ravishda 25, 16 va 4 ga teng bo'lsa, $GHKE$ to'rtburchak yuzini toping.

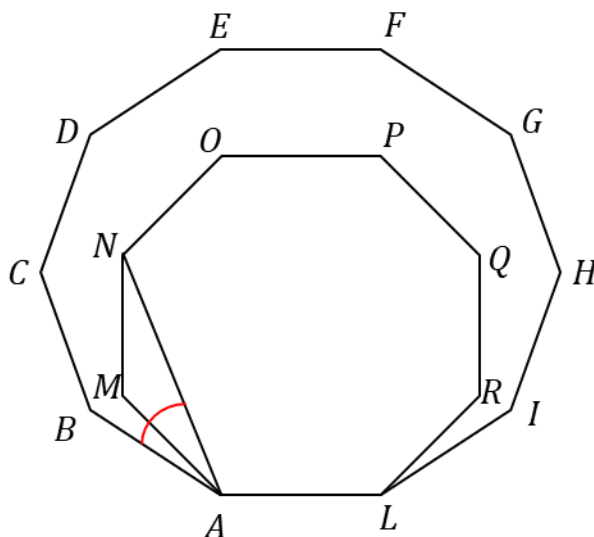


- A) 6
- B) 9
- C) 12
- D) 15

27. $y = \sqrt{x^2 - 6x + 9} + \sqrt{x^2 + 8x + 16}$ funksiyaning eng kichik qiymatini toping.

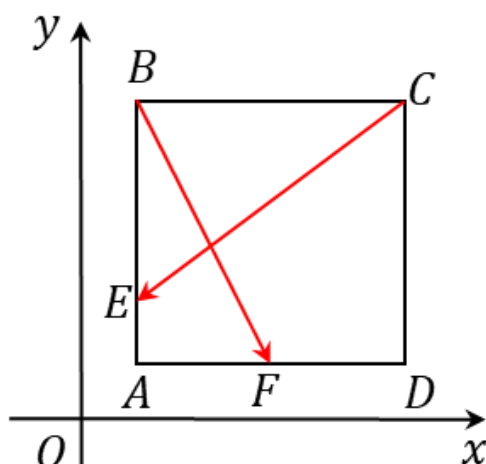
- A) 3
- B) 4
- C) 5
- D) 7

28. Rasmda $ABCDEFGHIL$ muntazam o'n burchak va $AMNOPQRL$ muntazam sakkizburchak tasvirlangan. Bunga ko'ra $\angle BAN$ burchakni toping.



- A) 9°
- B) $30,5^\circ$
- C) $31,5^\circ$
- D) $32,5^\circ$

29. Dekart koordinatalar sistemasida uchlari A, B, C va D nuqtalarda bo'lgan kvadrat berilgan (rasm). Agar $BC = 10$, $AE = 3$, $AF = 5$ bo'lsa, $\overrightarrow{CE} + \overrightarrow{BF}$ vektorning koordinatasini toping. ($AD \parallel Ox$).



- A) $(4; -8)$
- B) $(5; 14)$
- C) $(-5; -17)$
- D) $(5; -17)$

30. Tekislikdan a masofadan ikkita og'ma tushirildi. Ular tekislik bilan 45° li burchakni, o'zaro 60° li burchak hosil qiladi. Og'malarning uchlari orasidagi masofani toping.

- A) a
- B) $\sqrt{3}a$
- C) $\sqrt{2}a$
- D) $\frac{\sqrt{6}}{2}a$

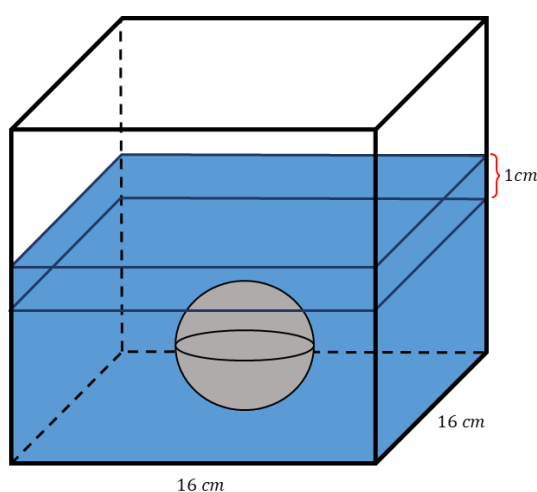
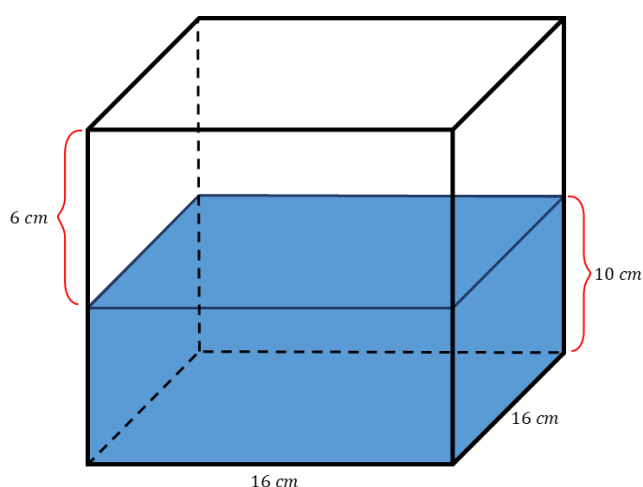
31. Agar $A = \{x | x = 4n + 3, n \in N\}$ va $B = \{x | x = 6n + 5, n \in N\}$ bo'lsa, $A \cap B$ ni toping.

- A) $\{x | x = 12n - 1, n \in N\}$
- B) $\{x | x = 12n - 11, n \in N\}$
- C) $\{x | x = 24n - 1, n \in N\}$
- D) $\{x | x = 24n - 11, n \in N\}$

32. Uch xonali sonlar ichida tanlangan son 5 ga bo'linish ehtimolligini toping.

- A) $\frac{1}{10}$
 B) $\frac{1}{20}$
 C) $\frac{1}{5}$
 D) $\frac{9}{50}$

Kub shaklidagi akvarium tomoni uzunligi 16 cm ga teng. Akvarium ichida asosdan hisoblaganda 10 cm balandlikda suv bor. Bu akvariumga metal shar solinganda, suv balandligi 1 cm ga ko'tarilgani ma'lum bo'lsa,



- A) 4
 B) 2
 C) 256
 D) 218
 E) $\frac{16}{3}$
 F) $\frac{8}{3}$

33. Shar radiusini toping(cm). ($\pi \approx 3$ deb oling)

34. Shar hajmini toping (cm^3).

35. Shar ichiga eng katta hajmli konus ichki chizilgan bo'lsa, konus balandligini toping(cm).

36. Tenglamani yeching:

$$(x^2 - 2024^2)^2 - 8096x - 1 = 0$$

a) Tenglama nechta haqiqiy ildizga ega?

Javob a) _____

b) Tenglamaning haqiqiy ildizlari yig'indisini hisoblang.

Javob b) _____

37. Tenglamani yeching:

$$\sin\left(x - \frac{\pi}{6}\right) + \sin\left(x - \frac{\pi}{3}\right) = \sin\left(x - \frac{\pi}{4}\right)$$

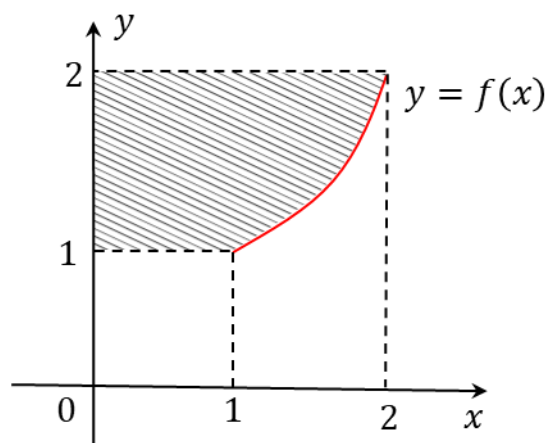
a) Tenglama $\left[-\frac{3\pi}{2}; \frac{\pi}{2}\right]$ oraliqta nechta yechimga ega?

Javob a) _____

b) Tenglama $\left[-\frac{3\pi}{2}; 0\right]$ oraliqdagi ildizlari yig'indisini toping.

Javob b) _____

38. Rasmda $f(x)$ funksiya grafigi tasvirlangan. Bo'yalgan soha yuzi $\frac{5}{3}$ ga teng bo'lsa,



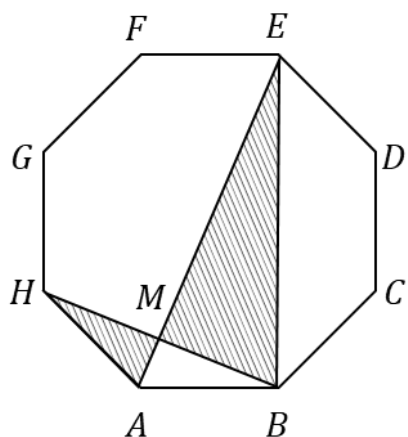
a) $S = \int_3^6 f\left(\frac{x}{3}\right) dx$ ni hisoblang.

Javob a) _____

b) $S^4 - 2S^2 + 5$ ni hisoblang.

Javob b) _____

39. $ABCDEFGH$ muntazam sakkizburchak. AE diagonal bilan HB diagonalalar M nuqtada kesishadi. HMA va BEM uchburchak yuzalar yig'indisi 10 ga teng bo'lsa,



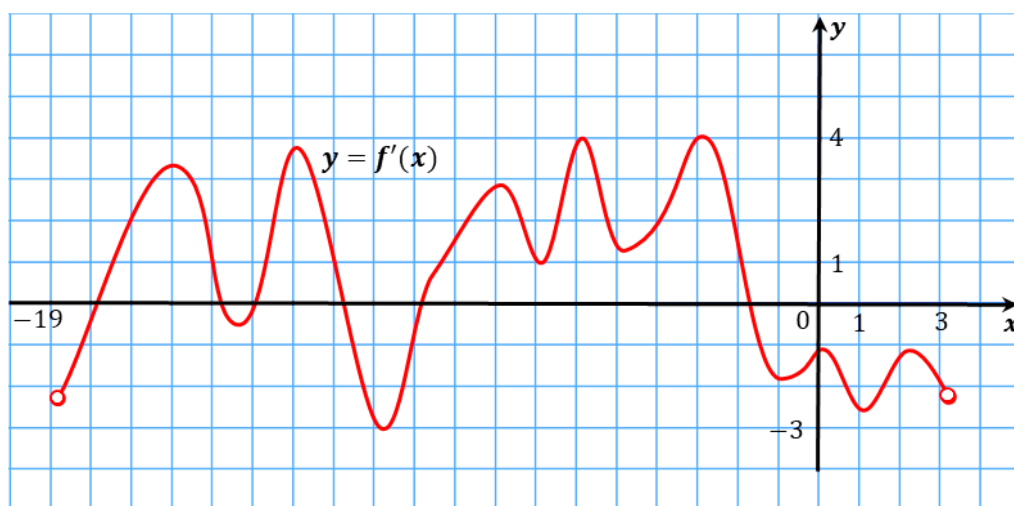
a) AE diagonal uzunligini toping

Javob a) _____

b) Muntazam sakkiz burchak yuzini toping.

Javob b) _____

40. Rasmda $(-19; 3)$ oraliqdagi $f(x)$ funksiyaning hosilasining grafigi tasvirlangan.



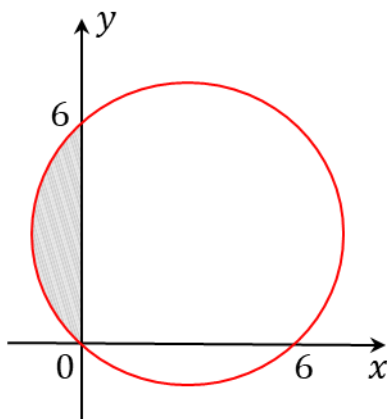
a) $f'(x) \cdot f''(x) = 0$ tenglamani qanoatlantiruvchi nechta butun son mavjud?

Javob a) _____

b) $f(x)$ funksiyaning nechta lokal maksimum nuqtasi mavjud?

Javob b) _____

41. Rasmda OY o'qini $(0; 6)$, OX o'qini $(6; 0)$ nuqtadan va koordinatalar boshidan o'tuvchi aylana berilgan.



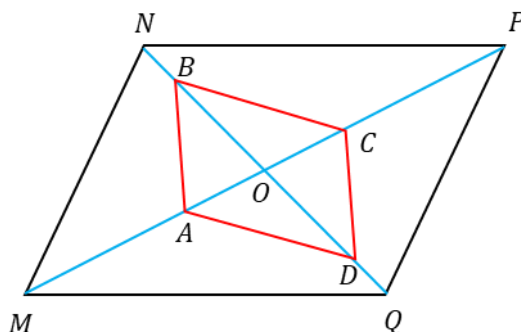
a) Bo'yalgan soha yuzini toping.

Javob a) _____

b) Bo'yalmagan soha yuzini toping.

Javob b) _____

42. $ABCD$ to'rtburchakning $MNPQ$ parallelogramning diagonallarida yotadi (rasm). $MNPQ$ parallelogramning yuzi 76 cm^2 ga teng. Agar $AC:MP = 3:8$ va $BD:NQ = 2:3$ bo'lsa,



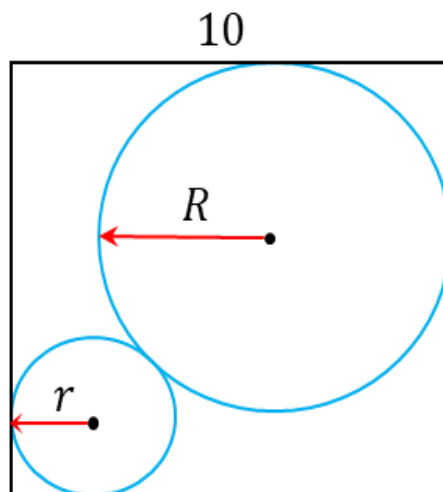
a) $ABCD$ to'rtburchakning yuzini (cm^2) toping.

Javob a) _____

b) $S_{MNPQ} - S_{ABCD}$ ni toping (cm^2).

Javob b) _____

43. Kvadratning tomoni 10 ga teng. $r = 16 - 10\sqrt{2}$ bo'lsa,



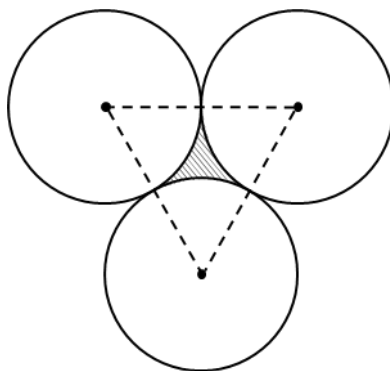
a) R ni toping.

Javob a) _____

b) Katta aylana uzunligini toping.

Javob b) _____

44. Rasm bir-biriga tegib turgan uchta bir xil doiradan iborat. Agar bo'yalgan soha yuzi $64\sqrt{3} - 32\pi$ bo'lsa,



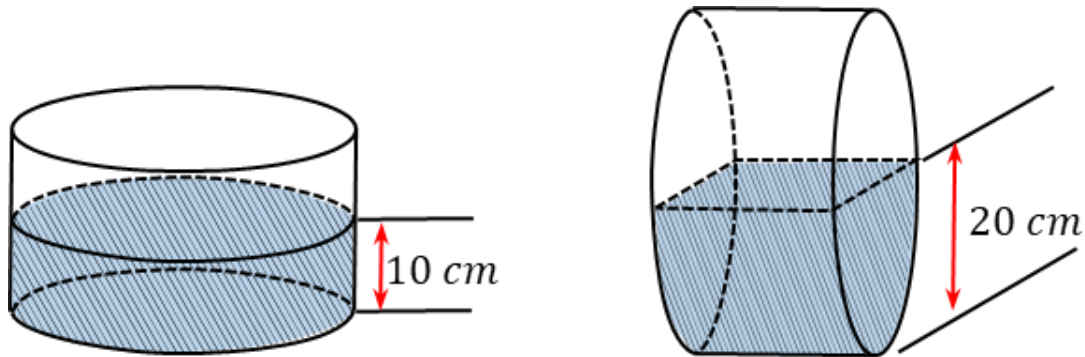
a) Har bir doira radiusini qancha?

Javob a) _____

b) Har bir doira yuzi nechaga teng?

Javob b) _____

45. Rasmlardagi ikkita bir xil to'g'ri dumaloq silindr idishlarning $\frac{1}{2}$ qismicha suyuqlik bor. Agar suyuqlik idish tagiga quyilgan bo'lsa, unda suyuqlikning chuqurligi 10 santimetr va agar idish yon tomoniga quyilgan bo'lsa, unda suyuqlikning chuqurligi 20 santimetr.



a) Silindrga eng kamida qancha cm^3 suv quysa to'ladi?

Javob a) _____

b) Idishda necha kub santimetr suyuqlik bor?

Javob b) _____